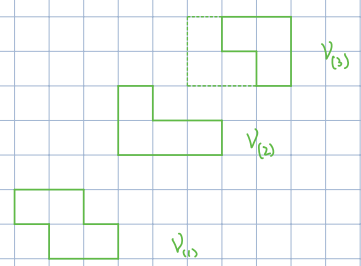
(See §4.1 in Path)

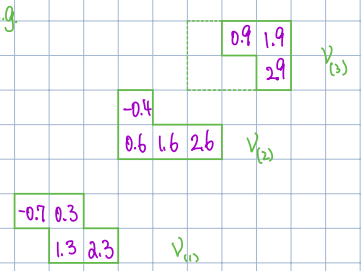
Let $\nu = (\nu_{(1)}, \dots, \nu_{(k)})$ be a tuple of skew diagrams. We consider the set of boxes in ν to be the disjoint union of the sets of boxes in the $\nu_{(i)}$.

e.g. $k=3$, $\nu = (32/1, 31, 33/21)$



Define the adjusted content of a box $a \in \nu_{(i)}$ to be $\tilde{c}(a) = c(a) + i\varepsilon$ ($\varepsilon > 0$ is fixed s.t. $k\varepsilon < 1$.)

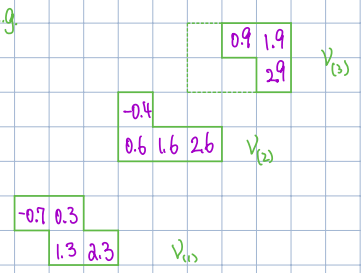
e.g.



$\tilde{c}(a)$ in each box

A diagonal in ν is the set of boxes of a fixed adjusted content, i.e. a diagonal of fixed content in one of the $\nu_{(i)}$.

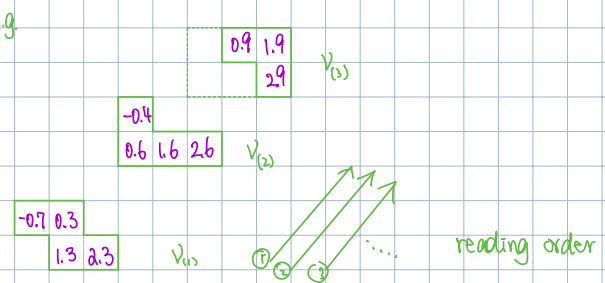
e.g.



There are 10 diagonals in ν, each consists of exactly one box in this case.

The reading order on ν is the total ordering $<$ on the boxes of ν s.t. $a < b \Rightarrow \tilde{c}(a) \leq \tilde{c}(b)$ and boxes on each diagonal (i.e. must be in the same shape) increases from SE to NE.

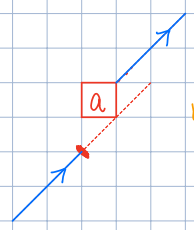
e.g.



reading order

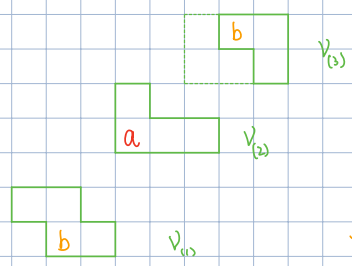
$\therefore a \in \nu_{(i)}, b \in \nu_{(j)} \Rightarrow i \neq j$ (i.e. different shape in ν)

An attacking pair (a, b) in ν: $a < b$ in reading order with $0 < \tilde{c}(b) - \tilde{c}(a) < 1$.



b = all boxes in the "cycle" and NOT in the same shape

e.g.



For this fixed a, there are two possible b's s.t. (a, b) is an attacking pair.

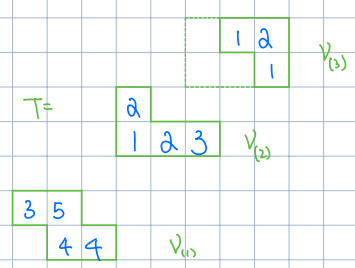
(a, b) is an attacking pair

A semistandard tableau on ν is a map $T: \nu \to \mathbb{Z}_+$ which restricts to a SSYT on each component ν_i.

SSYT(ν) = set of all semistandard tableau on ν

e.g. $\nu = (32/1, 31, 33/21)$

T=



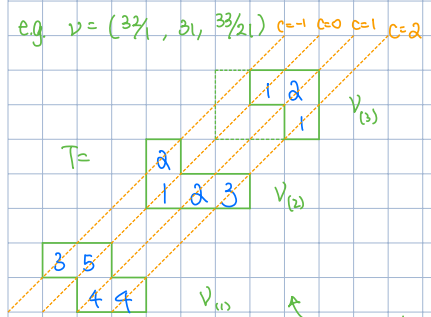
$\in$ SSYT(ν)

An attacking inversion in T = attacking pair (a, b) s.t. T(a) > T(b).

inv(T) = # attacking inversions in T.

e.g. $\nu = (32/1, 31, 33/21)$

T=



content of a

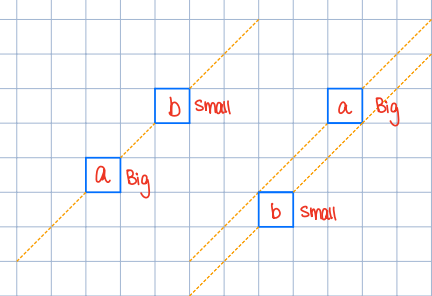
Inversions:

c=-1: (3,2)

c=0: (5,1), (5,11)

c=1: (4,2), (4,2),

c=2: (4,3), (4,1), (3,1)



inv(T) = 8

corresponds to $q^8 x_1^3 x_2^3 x_3^2 x_4^2 x_5$ in $\mathcal{G}_{(32/1, 31, 33/21)}$

Def: The Combinatorial LLT polynomial indexed by a tuple of skew Young diagrams ν is the generating function

$$\mathcal{G}_\nu(x; q) = \sum_{T \in SSYT(\nu)} q^{inv(T)} x^T$$

(see e.g. above)

where $x^T = \prod_{a \in \nu} x_{T(a)}$.

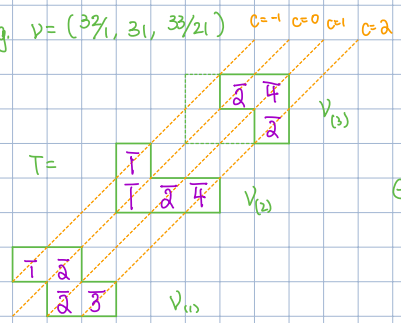
* $\mathcal{G}_\nu$ is known to be symmetric.

Fix an ordered alphabet $\mathcal{A}_-$ of 'negative' letters $\bar{1} < \bar{2} < \cdots$ (since $\mathcal{G}_\nu$ is symmetric, the choice of ordering is arbitrary)

A negative tableau on ν is a map $T: \nu \to \mathcal{A}_-$ that is strictly increasing on rows and weakly increasing on columns

$SSYT_-(\nu) :=$ set of all negative tableaux on ν.

eg. $\nu = (32/1, 31, 33/21)$

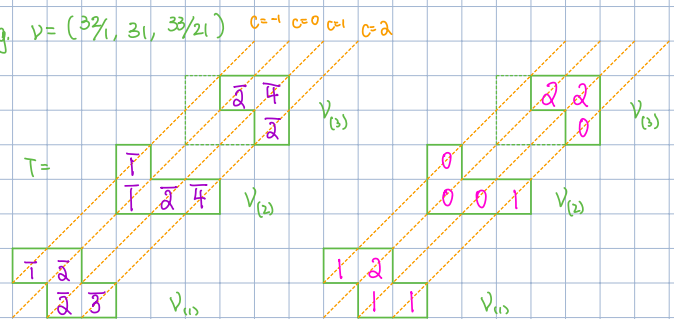


$\in SSYT_-(\nu)$

An attacking inversion in $T \in SSYT_-(\nu)$ = attacking pair (a,b) s.t. $T(a) \geq T(b)$.

$inv(T) = \#$ attacking inversions in T. (same as before)

eg. $\nu = (32/1, 31, 33/21)$



In each box a, the number in a is # boxes b s.t. (a,b) form an attacking inversion.

$\therefore inv(T) = 10$

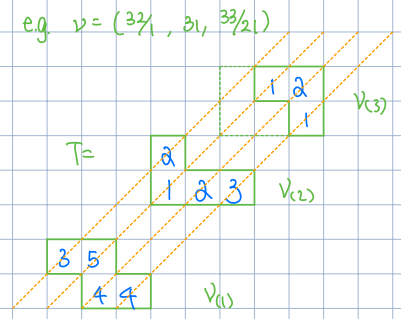
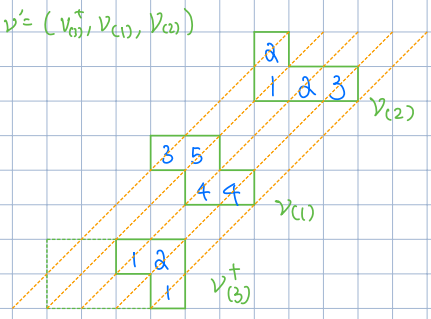
Prop 2.2.2 (Path: Corollary 4.1.3). Setting $x_{\bar{i}} = x_i \quad \forall \bar{i} \in \mathcal{A}_-$, we have

$$\omega \mathcal{G}_\nu(X;q) = \sum_{T \in SSYT_-(\nu)} q^{inv(T)} x^T.$$

Corollary 2.2.3: (Path Lemma 4.1.6)

The LLT polynomial $\mathcal{G}_\nu(X;q)$ is a linear combination of s_λ with $\ell(\lambda) \leq$ total no. of rows in $\nu_{(i)}$'s.

Prop. 2.2.4: Let $\nu = (\nu_{(1)}, \ldots, \nu_{(k)})$ and set $\nu' = (\nu^+_{(j+1)}, \ldots, \nu^+_{(k)}, \nu_{(1)}, \ldots, \nu_{(j)})$ for some $1 \leq j \leq k-1$, where $\nu^+_{(i)}$ is a translation of $\nu_{(i)}$ s.t. the content of every box is increased by 1. Then $\mathcal{G}_\nu(X;q) = \mathcal{G}_{\nu'}(X;q)$.

Proof: The construction of ν' gives a natural bijection between boxes of ν and boxes of ν' that preserves the reading order and the set of attacking pairs. This induces a bijection $SSYT(\nu) \cong SSYT(\nu')$ that preserves x^T and $inv(T)$. $\square$

e.g. $\nu = (32/1, 31, 33/21)$ $\quad j = 2 \quad \nu' = (\nu^+_{(3)}, \nu_{(1)}, \nu_{(2)})$